Viewpoint 4 Corrigendum

Cercis-Fisher Equivalence: Conditional Equivalence, Not Natural Emergence

SCX

July 2, 2026

1 Review Verdict

Original Claim	Verdict
$\text{Cercis}^2 \propto$ average KL divergence = squared Fisher geodesic length. "Naturally emerges" from information geometry.	Mathematically correct but "natural emergence" is overstated

1.1 Specific Issues

(1) **Not "average KL" but "minimum KL".** Cercis is the *minimum* KL divergence to the pure-gauge submanifold: $\mathcal{C} \approx 2 \cdot \min_h \mathrm{KL}(P_A \| P_{d_0 h})$, not an "average KL divergence."

(2) **Not "natural emergence" but conditional equivalence.** The $\text{GeoDist}^2 \leftrightarrow 2\text{KL}$ equivalence depends on 8 strong assumptions (see §2). If any single assumption fails, the equivalence breaks. SCX data is extremely unlikely to form an exponential family.

(3) **Complete breakdown outside exponential families.** For general statistical manifolds, the difference between GeoDist^2 and 2KL is controlled by the Amari-Chentsov third-order tensor. This tensor is generically non-zero for SCX data, so the equivalence degrades.

2 The Eight Assumptions

For $\text{GeoDist}(p,q)^2 = 2\mathrm{KL}(p\|q) + O(\|p-q\|^3)$ to hold rigorously, the following eight assumptions must be simultaneously satisfied. Failure of any single one degrades the equality.

H1: Exponential Family Assumption.

The output distribution family $\mathcal{P} = \{p(x \mid \theta)\}$ must be an exponential family, i.e., there exists a sufficient statistic $T(x)$ and cumulant generating function $\psi(\theta)$ such that $p(x \mid \theta) = h(x)\exp(\langle \theta, T(x)\rangle - \psi(\theta))$. Only under this condition is the Fisher information matrix the Hessian of ψ, and KL divergence equals Bregman divergence (Lemma 4.3 / Lemma **??** in the original context).

Consequence of failure: Outside exponential families, $\mathrm{KL}(\theta_1\|\theta_2)$ is no longer a Bregman divergence of ψ, and the second-order term of the Taylor expansion is no longer $g_{ij}\Delta\theta^i\Delta\theta^j$. The difference between GeoDist^2 and 2KL is controlled by the Amari-Chentsov third-order tensor (Open Problem 6.2 / Open Problem **??**).

H2: Small-Distance / Local Approximation Assumption.

θ_p and θ_q must be sufficiently close so that third-order and higher remainders $O(\|\theta_q - \theta_p\|^3)$ are negligible. All derivations truncate the Taylor expansion at second order.

Consequence of failure: For large discrepancies (common in cross-domain SCX experts), third-order error terms dominate and $\text{GeoDist}^2 \approx 2\text{KL}$ no longer holds.

H3: C^2 **Differentiability Assumption.**
The map $\theta \mapsto p(x \mid \theta)$ must be C^2 for almost all x. This is required for second-order Taylor expansion and for the Fisher matrix as a second derivative.

Consequence of failure: If the density has non-differentiable points (e.g., piecewise distributions from ReLU activations), the Fisher metric is undefined at those points and geodesics are not uniquely determined.

H4: Integration-Differentiation Interchangeability Assumption.
The interchange $\partial_\theta \int p(x|\theta)d\mu(x) = \int \partial_\theta p(x|\theta)d\mu(x)$ must hold. This is the standard regularity condition ensuring $\mathbb{E}[\partial_\theta \log p] = 0$, which underpins the Fisher information definition.

Consequence of failure: If the support depends on θ (e.g., uniform distributions), the standard Fisher information form breaks down.

H5: Fisher Full-Rank Assumption.
The Fisher information matrix $\mathcal{I}(\theta)$ must be positive-definite (full rank) at every point. This is required for (Θ, g) to be a genuine Riemannian manifold — a degenerate metric does not define unique geodesics.

Consequence of failure: If parameters are functionally dependent (e.g., normalization constraints causing effective dimensionality reduction), the Fisher matrix is singular, geodesics may not exist or be unique, and GeoDist is undefined.

H6: Small-Connection Approximation Assumption.
Edge assignments a_e and gauge transformations h_v must be within the small-connection limit ($\max_e \mathrm{dist}_{\mathrm{O}(d)}(A_e, I_d) < r_0$, the matrix logarithm convergence radius). Only then does non-abelian gauge-fixing reduce to linear Hodge theory on $\mathfrak{so}(d)$.

Consequence of failure: Under large rotations (e.g., 90° frame misalignment between cross-modal experts), nonlinear Cartan terms cannot be neglected, and the relationship between GeoDist and Euclidean distance breaks.

H7: Euclidean Approximation / Degenerate Fisher Assumption.
For the Euclidean Cercis definition ($\|r\|^2 = \|r_{\mathrm{harm}}\|^2 + \|r_{\mathrm{coexact}}\|^2$) to coincide with the Fisher definition, the Fisher metric must degenerate to Euclidean: $\mathcal{I}_{ij}(\theta) = \delta_{ij}$. This means all directions on the information manifold have equal weight and zero curvature.

Consequence of failure: Real SCX parameter spaces almost certainly have non-trivial Fisher curvature. A non-negligible curvature correction exists between Euclidean Cercis and Fisher geodesic distance.

H8: Distribution Compactness / Bounded Support Assumption.
All expert output distributions must be sufficiently concentrated on a compact subset of Θ, so that the local Taylor expansion is uniformly valid across all boundary observations and gauge bulk states.

Consequence of failure: Heavy-tailed distributions (e.g., Student's-t) may place some observations outside the local validity domain, causing uncontrolled growth of third-order error terms on individual edges.

3 Corrected Claim

3.1 English Correction

Original claim (overstated, to be removed):

> "The Cercis Score is the geodesic distance under the Fisher information metric. Under the exponential family assumption, Cercis^2 $\propto$ mean KL divergence = squared Fisher geodesic length. The audit score is not artificially defined — it naturally emerges from the information geometry of the data manifold."

Corrected claim (conditional equivalence):

> **Under the exponential family assumption and local approximation,** the Cercis Score equals twice the **minimum** KL divergence from the observed edge distribution to the pure-gauge submanifold, which in turn equals (to second order) the squared Fisher geodesic distance. This provides an information-geometric **interpretation** — not a "natural emergence" but a **conditional equivalence** that holds when and only when the Situs manifold admits an exponential family structure. **For non-exponential-family data, the relationship degrades by the Amari-Chentsov tensor (Open Problem 6.2).**

Explicit formulation:

Theorem (Information-Geometric Conditional Equivalence of Cercis, Revised)
Assumptions: All 8 assumptions (H1–H8) hold simultaneously.
Conclusion:

$$\mathcal{C} = \min_{d_0 h \in \mathrm{im}(d_0)} \mathrm{GeoDist}(P_A,\ P_{d_0 h})^2 = 2 \min_{d_0 h \in \mathrm{im}(d_0)} \mathrm{KL}(P_A \| P_{d_0 h}) + O(\|\mathfrak{a}\|^3).$$

Cercis measures the minimal Fisher information distance from the observed expert edge distributions to the "pure-gauge bulk states" — the hypothetical world where all experts can be perfectly aligned.
Limitations:

(i) The equivalence is rigorous within exponential families (H1); outside them, the difference between $\mathrm{GeoDist}^2$ and 2KL is controlled by the Amari-Chentsov third-order tensor $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$, which is generically non-zero for SCX data (Open Problem **??**).

(ii) The equivalence holds under local approximation (H2); for large discrepancies, third-order terms dominate.

(iii) "Natural emergence" is an incorrect characterization — this is a *constructed equivalence* under strong assumptions, not an intrinsic geometric inevitability of the data.

4 Five Breakdown Points

Below, we summarize five principal ways the equivalence can break on real SCX data:

B1: Non-Exponential-Family Failure (Most Severe).
SCX expert outputs (MoE routing weights, softmax scores, attention distributions) are extremely unlikely to form an exponential family. Exponential families require specific algebraic structure (linear sufficient statistics $T(x)$) that deep learning outputs do not generally satisfy. The document itself lists this as Open Problem 6.2 (**??**).

B2: Confusing "min KL" with "average KL".
Cercis solves $\min_h \mathrm{KL}(P_A \| P_{d_0 h})$ — a minimization problem, not an average of KL divergences over all expert pairs. These are mathematically distinct.

B3: Hidden Euclidean Approximation Requirement.
The equivalence between Euclidean-norm Cercis and Fisher geodesic distance requires $\mathcal{I}_{ij} = \delta_{ij}$ (H7) — meaning the information manifold must be everywhere flat, which is extremely unlikely on real SCX parameter spaces.

B4: Intrinsic Correction in Theorem 4.2 Proof.
Choosing the wrong submanifold loses the harmonic component. The correct definition is: the bulk submanifold is parameterized by exact connections (pure gauge) $\mathrm{im}(d_0)$, not harmonic bulk states $\ker(L_1)$. Theorem **??** identifies and corrects this issue in its proof.

B5: Locality Constraint.
All results hold only when $\|\theta_q - \theta_p\|$ is sufficiently small. When inter-configuration expert differences are large (e.g., different domains, different training stages), the third-order error term $O(\|\Delta\theta\|^3)$ is non-negligible.

5 Recommended Revisions

5.1 Changes Needed in gauge_formalized.tex

(1) **Abstract (lines 234–238):** Remove "natural emergence" language. Change

> "We prove that under this family, Fisher geodesic length and KL divergence satisfy local equivalence..."

to:

> "We prove that under the exponential family assumption and local approximation, a conditional equivalence relation holds between Fisher geodesic length and KL divergence..."

Modify the English abstract similarly.

(2) **Section ?? title:** Change

> "Fisher Geodesic–KL Equivalence under Exponential Families"

to:

> "Conditional Fisher Geodesic–KL Equivalence under Exponential Families"

(3) **Theorem ??:** Add an explicit list of assumptions before the theorem statement, referencing H1–H8 above. Add a limiting remark at the end of the theorem.

(4) **Theorem ?? (Unified Structure Theorem):** Change the phrasing from

> "Cercis is reformulated as the Fisher geodesic distance from boundary to pure-gauge bulk states, locally equivalent to 2KL under exponential families"

to:

> "Under the exponential family assumption and local approximation, Cercis can be reinterpreted as the Fisher geodesic distance from boundary to pure-gauge bulk states, conditionally equivalent to 2KL (to second order). Behavior outside exponential families is handled by Open Problem 6.2."

(5) **Comparison table (lines 1529–1532):** For

> "Cercis definition: $\min_{d_0 h} \mathrm{GeoDist}(P_A, P_{d_0 h})^2$" and "Cercis probabilistic interpretation: $\approx 2 \min_{d_0 h} \mathrm{KL}(P_A \| P_{d_0 h})$"

add a footnote or parenthetical note: "(requires exponential family assumption; degrades outside exponential families)"

(6) **Section ?? "Rigor of Information Geometry" paragraph (lines 1560–1564):** Change

> "Its application to SCX's $\mathcal{S}$ manifold in this paper is a natural generalization..."

to:

> "Its application to SCX's $\mathcal{S}$ manifold in this paper is a conditional generalization under specific assumptions. When these assumptions are not satisfied (especially the non-exponential-family case, Open Problem 6.2), the equivalence degrades via the Amari-Chentsov tensor."

(7) **Conclusion item 3 (lines 1678–1684):** Change

> "proved the local equivalence of Fisher geodesic length and KL divergence under exponential families... established an information-geometric interpretation of the Cercis Score"

to:

> "proved the conditional equivalence of Fisher geodesic length and KL divergence under the exponential family assumption... established a conditional information-geometric interpretation of the Cercis Score, explicitly noting the 8 prerequisite assumptions and failure conditions."

6 Cross-Reference to Open Problem 6.2

This corrigendum is directly linked to Open Problem **??** ("Fisher-KL Equivalence Outside Exponential Families"), which states:

> For general SCX output distributions (potentially non-exponential-family), the relationship between Fisher geodesic distance and KL divergence is controlled by the Amari-Chentsov third-order tensor $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$. Bounds on this tensor for empirical SCX output distributions are needed.

Review basis: This corrigendum is based on: `docs/reviews/GAUGE_5VIEWPOINTS_FINAL.md` (verdict: "overstated") and `docs/reviews/GAUGE_VIEWPOINTS_REVIEW.md` (detailed analysis). The verdict on Viewpoint 4 is: ✓ mathematically correct relationship but "natural emergence" is severely overstated — downgrade to "conditional equivalence".